DA 5001/6336 (Jan-May 2026): HW2

IIT Madras

Due Date: March 6, 2026 at 11:59 PM

Instructions

- Theory questions are not graded, so submission of the solution is optional. We strongly recommend that you work them out to fully understand the material (which will also be tested via the mini-quizzes).
- The simulation exercises are graded. (Note that this HW does not have any simulation exercises.)
- List the names of students you collaborated with for the homework. Also, provide citations to any books, notes, or web resources you used for any problem. Failure to do so will be considered as a violation of the honour code.
- If you used an LLM for help in one of the exercises, you must specify the name of the LLM and the exact prompt used.

Honour Code

Please read the full honour code on the course webpage and write "I ACCEPT THE HONOUR CODE" in your submission. Your submission will not be graded if you do not accept the honour code.

1 Alternate Definitions of DP That Do Not Work (10 marks)

In this exercise, we will investigate the pitfalls of common statistical divergences like the KL divergence (natural choice for ML folks) or the total variation distance (natural for probabilists, cryptographers, etc.) as notions of privacy. **Note**: this exercise is conceptually challenging (in contrast, other problems are conceptually easy but might involve detailed derivations).

Total Variation (TV) Distance Recall that the total variation distance between two probability distributions P, Q over a space $\mathcal{Y}$ is defined as

$$\mathrm{TV}(P, Q) = \sup_{S \subset \mathcal{Y}} \{P(S) - Q(S)\} \ .$$

It can be shown that $\mathrm{TV}(P, Q) = \frac{1}{2} \int_{\mathcal{Y}} |f_P(y) - f_Q(y)| \, dy$ for continuous P, Q with densities f_P, f_Q, while for discrete distributions, an analogous expression holds where the integral is replaced by the sum and the densities by the probability mass functions.

The TV distance quantifies an additive error in the probabilities of any outcome, while ε-DP quantifies the multiplicative error. Suppose we define an alternative to DP based on the TV distance.

Definition 1. *We say that a randomized algorithm $\mathcal{A}(\cdot)$ is δ-TV private if* $\mathrm{TV}\big(\mathcal{A}(D), \mathcal{A}(D')\big) \leq \delta$ *for any adjacent datasets $D \simeq D'$.*

Let us consider what goes wrong with such a definition.

1. If a randomized algorithm $\mathcal{A}(\cdot)$ is ε-DP, then show that it is δ-TV private with $\delta = 1 - e^{-\varepsilon}$.
2. Consider the "replace-one" notion of adjacency (e.g.that we considered with randomized response). That is, $D \simeq D'$ are considered adjacent if we can obtain D' by substituting one of the elements of D so that $|D \cap D'| + 1 = |D| = |D'|$.
   Consider a randomized algorithm $\mathcal{A}_0 : \mathcal{X}^n \to \mathcal{X}$ such that $\mathcal{A}_0(x_1, \ldots, x_n)$ releases one of its inputs uniformly at random.[1] Prove that $\mathcal{A}_0$ is δ-TV private with $\delta = 1/n$.

Any algorithm that releases one datapoint is not what we would intuitively think of as "private". Thus, our notion of δ-TV privacy does not make sense of δ around $1/n$ or larger. Let us consider what happens when $\delta < 1/2n$ is small. For this, we will need some additional properties of TV.

3. Prove that TV is a *distance metric* (in the mathematical sense of the term). That is, prove that it satisfies:
   (a) Symmetry: $\mathrm{TV}(P, Q) = \mathrm{TV}(Q, P)$.
   (b) Non-negativity: $\mathrm{TV}(P, Q) \geq 0$ with $\mathrm{TV}(P, Q) = 0$ if and only if $P = Q$.
   (c) Triangle Inequality: $\mathrm{TV}(P, Q) \leq \mathrm{TV}(P, S) + \mathrm{TV}(S, Q)$ for any probability distributions P, Q, S.
4. Suppose that $\mathcal{A}(\cdot)$ satisfies δ-TV privacy. Then, show that for any *arbitrary pairs of datasets* $D, D' \in \mathcal{X}^n$ (that are not required to be adjacent), we have $\mathrm{TV}\big(\mathcal{A}(D), \mathcal{A}(D')\big) \leq n\delta$. **Hint**.[2]

Here is why δ-TV privacy with small δ fails. If $\delta < 1/2n$, then we have that $\mathrm{TV}\big(\mathcal{A}(D), \mathcal{A}(D')\big) \leq 1/2$. That is, with probability at least $1/2$, the output $\mathcal{A}(D)$ is independent of D (since the output should match that of any arbitrary dataset D'). In other words, such a randomized algorithm is useless. Thus, the TV distance is not a good metric for defining differential privacy (i.e. between adjacent datasets).

Note, however, that (ε, δ)-DP is a combination of δ-TV with ε-DP, so the TV distance is not entirely unusable for privacy. Comparing the definition of the hockey-stick divergence (see Problem 5) with that of TV distance makes the connection apparent.

KL Divergence We will now see how the KL divergence is an unsuitable notion of differential privacy because it is not sensitive enough to low probability bad events.

5. Construct four discrete distributions P_1, Q_2, P_2, Q_2 with a support size of k of your choice such that their privacy loss random variables $Z_1 \sim \mathrm{PrivLoss}(P_1, Q_1)$ and $Z_2 \sim \mathrm{PrivLoss}(P_2, Q_2)$ satisfy the following:
   (a) $\mathbb{E}[Z_1] = \mathbb{E}[Z_2]$ or equivalently, we have $\mathrm{KL}(P_1, Q_1) = \mathrm{KL}(P_2, Q_2)$.
   (b) For some $\varepsilon > 0$, we have $\mathbb{P}(Z_1 \leq \varepsilon) = 1$ and $\mathbb{P}(Z_2 > \varepsilon) \geq 0.1$.

Why is this bad from a privacy perspective?

[1] This is also known colloquially as the "name and shame" mechanism.

[2] **Hint**. Construct a series of datasets $D = D_0 \simeq D_1 \simeq \cdots \simeq D_k = D'$ for some $k \leq n$ by swapping out one element at a time. Then, apply the triangle inequality.

2 Properties of the Rényi Divergences and DP (10 marks)

Recall that the Rényi divergence of order $\alpha \in (0, \infty) \setminus \{1\}$ between two discrete distributions P and Q is defined as

$$R_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \left(\mathbb{E}_{X \sim P} \left(\frac{P(X)}{Q(X)} \right)^{\alpha - 1} \right) = \frac{1}{\alpha - 1} \log \left(\mathbb{E}_{X \sim Q} \left(\frac{P(X)}{Q(X)} \right)^{\alpha} \right) .$$

For continuous distributions P, Q with respective PDFs f_P, f_Q, the likelihood ratio $P(X)/Q(X)$ is replaced by the density ratio $f_P(X)/f_Q(X)$. The Rényi divergence for orders $\alpha = 0, 1, \infty$ is defined as the limit of the expression above.

We will establish a number of properties of the Rényi divergence. You may assume that $P = (P_1, \ldots, P_k)$ and $Q = (Q_1, \ldots, Q_k)$ are discrete distributions over k items unless specified otherwise.

1. Prove that $\lim_{\alpha \to 1} R_\alpha(P\|Q) = \mathrm{KL}(P\|Q)$ recovers the KL divergence. **Hint**.[3]

2. Prove that $R_\infty(P\|Q) := \lim_{\alpha \to \infty} R_\alpha(P\|Q) = \log \left(\max_{i=1,\ldots,k} \frac{p_i}{q_i} \right) = \max_{i=1,\ldots,k} \log \left(\frac{p_i}{q_i} \right)$. **Hint**.[4]
   Context: Note that this notion of divergence corresponds exactly to ε-DP: a randomized algorithm $\mathcal{A}(\cdot)$ is ε-DP iff $R_\infty\big(\mathcal{A}(D)\|\mathcal{A}(D')\big) \le \varepsilon$ for all adjacent datasets $D \simeq D'$.

3. Prove that $R_\alpha(P\|Q)$ is a non-decreasing function of $\alpha > 1$ for any pair of fixed distributions P, Q. **Hint**.[5]
   Context: As α gets larger, $R_\alpha(P\|Q)$ focuses more on tail events, i.e. worst-case sets S where $P(S)$ is large and $Q(S)$ is small. Indeed, $\alpha \to 1$ corresponds to the KL divergence, which is not sensitive to the tails (Problem 1), while $\alpha \to \infty$ recovers ε-DP which is overly sensitive to low-probability tail events. Rényi divergence is flexible enough to focus the analysis on intermediate values of α where we can capture the best of both worlds.

4. Alternate form of concentrated DP: Prove that a randomized algorithm $\mathcal{A}(\cdot)$ is ρ-zCDP iff
   $$R_\alpha\big(\mathcal{A}(D)\|\mathcal{A}(D')\big) \le \rho\alpha$$
   for all $\alpha > 1$ and for all adjacent datasets $D \simeq D'$.

5. Derive from first principles the following formula for d-dimensional Gaussians with a common covariance matrix $\Sigma \in \mathbb{S}^d_{++}$:
   $$R_\alpha\big(\mathcal{N}(\mu, \Sigma)\|\mathcal{N}(\mu', \Sigma)\big) = \frac{\alpha}{2} \, (\mu - \mu')^\top \Sigma^{-1} (\mu - \mu') \, .$$
   Context: This linear scaling of the α-Rényi divergence with α for Gaussians has some direct implications. First, this directly implies the zCDP property of the Gaussian mechanism. Second, the behaviour is qualitatively the same for all α, i.e., zCDP fully captures all properties of the Gaussian mechanism. As we see in class, subsampling plus composition leads to non-linear behavior with α, where it makes sense to focus on intermediate values of α.

[3] **Hint**: Use L'Hôpital's rule.

[4] **Hint**: How do you establish that the vector p-norms satisfy $\|v\|_\infty := \lim_{p \to \infty} \|v\|_p = \max_i |v_i|$? Try the same ideas.

[5] **Hint**: There are many ways to prove this. One option is to use Jensen's inequality on the convex function $f(t) = t^{\frac{\alpha' - 1}{\alpha - 1}}$ for $1 < \alpha < \alpha' < \infty$ to show that

$$\exp\big((\alpha' - 1) R_\alpha(P\|Q)\big) \le \exp\big((\alpha' - 1) R_{\alpha'}(P\|Q)\big) \, .$$

An alternate option is to define $h(\alpha) := R_\alpha(P\|Q)$ (with arbitrary P, Q fixed), note that it is continuous and differentiable and show that its derivative $h'(\alpha)$ is non-negative.

3 From Rényi Divergences to Properties of DP (10 marks)

We will continue establishing properties of the Rényi divergence from the previous exercise. These properties directly imply desirable properties of DP variants such as Rényi DP and zCDP.

1. Prove that $R_\alpha(P_1 \times P_2 \| Q_1 \times Q_2) = R_\alpha(P_1 \| Q_1) + R_\alpha(P_2 \| Q_2)$. Recall that $P_1 \times P_2$ denotes the product distribution of P_1 and P_2; a sample from $P_1 \times P_2$ can be obtained by sampling $X_1 \sim P_1$, $X_2 \sim P_2$ and returning (X_1, X_2).
   Context: This leads to a composition result for Rényi DP and hence, zCDP as well. Note that this is identical to the zCDP composition result we proved in class.

2. Prove the triangle-like inequality: for all $1 < \alpha < \alpha' < \infty$ and distributions P, Q, S with $R_\infty(S \| Q) < \infty$, we have
$$R_\alpha(P \| Q) \leq \frac{\alpha'}{\alpha' - 1} R_\beta(P \| S) + R_{\alpha'}(S \| Q)\,,$$
   where $\beta = \frac{\alpha(\alpha'-1)}{\alpha'-\alpha}$. You may use Hölder's inequality which states that
$$\mathbb{E}[XY] \leq \left(\mathbb{E}[X^p]\right)^{1/p} \left(\mathbb{E}[Y^q]\right)^{1/q}$$
   for all random variables X, Y (such that the expectations above are well-defined) and all $p, q > 1$ satisfying $1/p + 1/q = 1$. **Hint**.[6]

3. Prove the triangle inequality for zCDP: if $R_\alpha(P \| S) \leq \rho_1 \alpha$ and $R_\alpha(S \| Q) \leq \rho_2 \alpha$ for all $\alpha > 1$, then we have
$$R_\alpha(P \| Q) \leq \left(\sqrt{\rho_1} + \sqrt{\rho_2}\right)^2 \alpha \quad \forall \alpha > 1.$$
   Hint.[7]
   Context: This triangle inequality can be used to prove a property known as group privacy. If $D \simeq D'$ and $D' \simeq D''$ are two pairs of adjacent datasets, then D and D'' differ by two elements. But, the zCDP property can be extended to hold for all such pairs D, D'' using the triangle inequality above. In general, this can be extended to pairs of datasets that differ by k elements and is known as "group privacy". This is because the algorithm is now private w.r.t. changes in *groups* of k datapoints.

4 Sub-Exponential Concentration

Recall that a real-valued random variable Y is *sub-exponential* with parameters (σ^2, b) if
$$\log \mathbb{E}\left[e^{\lambda Y}\right] \leq \frac{\sigma^2 \lambda^2}{2} \qquad \text{for all } |\lambda| < \frac{1}{b}.$$

[6] **Hint**: Show using Hölder's inequality that for all $p, q > 1$ with $1/p + 1/q = 1$, we have
$$\exp\left((\alpha - 1) R_\alpha(P \| Q)\right) \leq \exp\left(\frac{\alpha p - 1}{p} R_{\alpha p}(P \| S)\right) \exp\left((\alpha - 1) R_{(\alpha-1)q+1}(S \| Q)\right)\,.$$
Now, rearrange and substitute appropriate values for p, q.

[7] **Hint**: Use the triangle-like inequality you just proved for Rényi divergences and find (using calculus) the value of α' that gives the smallest upper bound for each α.

In this problem, we establish sub-exponential concentration for the average of χ^2 random variables with one degree of freedom. Let $X_1, \ldots, X_n$ be i.i.d. random variables with distribution χ_1^2, i.e. $X_i^2 = Z_i^2$ for $Z_i \sim \mathcal{N}(0,1)$. Define

$$\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i,$$

with mean $\mu := \mathbb{E}[X_i] = 1$ We will derive a classic sub-exponential tail bound of the following form:

$$\bar{X} - 1 \leq O\left(\sqrt{\frac{\log(1/\delta)}{n}} + \frac{\log(1/\delta)}{n}\right),$$

holding with probability at least $1 - \delta$.

Recall that you also showed

$$\mathbb{E}[e^{\lambda(X-1)}] = \frac{1}{\sqrt{1-2\lambda}} \quad \text{for } \lambda < 1/2.$$

1. Identify explicit constants (σ^2, b) such that

$$\log \mathbb{E}\Big[e^{\lambda(X-1)}\Big] \leq \frac{\sigma^2\lambda^2}{2} \qquad \text{for all } 0 \leq \lambda < \frac{1}{b}.$$

   Hint.[8]

2. Show that for $\lambda \in (0, \lambda_{\max})$ (derive $\lambda_{\max}$ explicitly):

$$\log \mathbb{E}\Big[e^{\lambda(\bar{X}-1)}\Big] \leq \frac{\sigma^2}{2n}\lambda^2.$$

3. Next, we consider a Chernoff bound for some fixed $t > 0$:

$$\mathbb{P}(\bar{X} - 1 \geq t) \leq \inf_{0<\lambda<\lambda_{\max}} \exp\Big(-\lambda t + \log \mathbb{E}e^{\lambda(\bar{X}-1)}\Big) .$$

   By using the log-MGF upper bound, this can be upper bounded by the minimum of a quadratic. Find its unconstrained optimum $\lambda_\star$. Argue that the optimal feasible λ from above should either be $\lambda_\star$ (interior optimum) or $\lambda_{\max}$ (boundary optimum).

4. From the above, conclude a tail bound of the form

$$\mathbb{P}(\bar{X} - 1 \geq t) \leq \exp\big(-n \min\{c_1 t^2, c_2 t\}\big)$$

   for some explicit constant $c_1, c_2 > 0$. Specify the exact constants.

5. Convert your tail bound into the following quantile statement: for any $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$\bar{X} \leq 1 + A\sqrt{\frac{\log(1/\delta)}{n}} + B\,\frac{\log(1/\delta)}{n},$$

   for explicit constants $A, B > 0$.

[8] **Hint**: Use the upper bound $-\log(1-u) \leq \frac{u}{1-u}$ for all $u \in (0,1)$. Then, restrict to a simple range such as $\lambda \in [0, 1/4)$ to obtain clean constants.

Note:

- The $\sqrt{\log(1/\delta)/n}$ term corresponds to the CLT regime. If you average enough chi-squared random variables (i.e. n is large), the average starts to behave like a Gaussian due to CLT.
- The $\log(1/\delta)/n$ term arises out of the maximum of the range $\lambda_{\max}$. When you take an extremely rare event far in the right tail (i.e. when $\log(1/\delta)$ is comparable to or larger than n), this term dominates. In this regime, you get a worse rate, which is the price of asking for ultra-high confidence beyond the CLT regime.

5 [Optional] Privacy Loss to DP Conversions (10 marks)

Consider two distributions P, Q over some set $\mathcal{Y}$ such that $\mathrm{PrivLoss}(P, Q)$ is well-defined. Define the hockey-stick divergence H_α as

$$H_\alpha(P\|Q) = \sup_{S\subset\mathcal{Y}} \left\{P(S) - \alpha\, Q(S)\right\}.$$

Note that $\alpha = 1$ is related to the total variation distance, which we explored further in Problem 1.

1. (ε, δ)-DP in terms of the hockey-stick divergence: Prove that a randomized algorithm $\mathcal{A}(\cdot)$ is (ε, δ)-DP iff $H_{e^\varepsilon}\big(\mathcal{A}(D)\|\mathcal{A}(D')\big) \le \delta$ for all adjacent datasets $D \simeq D'$.
2. Prove that

$$H_{e^\varepsilon}(P\|Q) = \mathbb{P}_{Z\sim\mathrm{PrivLoss}(P,Q)}(Z > \varepsilon) - e^\varepsilon\ \mathbb{P}_{Z'\sim\mathrm{PrivLoss}(Q,P)}(Z' < -\varepsilon)\,.$$

   Hint.[9]

3. Compute the hockey-stick divergence $H_{e^\varepsilon}(\mathcal{N}(0,1), \mathcal{N}(\Delta, 1))$ using the expression above. You can express your answer in terms of the standard normal tail probability $\Phi(t) = \mathbb{P}\big(\mathcal{N}(0,1) > t\big)$.
4. Consider a deterministic function $A : \mathcal{X}^* \to \mathbb{R}$ with sensitivity $\Delta = \max_{D\simeq D'} |A(D) - A(D')|$. Then, the randomized algorithm $\mathcal{A}(D) = \mathcal{N}\big(A(D), \sigma^2\big)$ satisfies (ε, δ)-DP with

$$\delta = \delta(\varepsilon) = \Phi\left(\frac{\varepsilon - \rho}{\sqrt{2\rho}}\right) - e^\varepsilon\ \Phi\left(\frac{\varepsilon + \rho}{\sqrt{2\rho}}\right),$$

   where $\rho = \frac{\Delta^2}{2\sigma^2}$ and $\Phi(t) = \mathbb{P}\big(\mathcal{N}(0,1) > t\big)$ is the standard normal tail probability.

5. Set $\Delta = 1$ and $\sigma = 1$. Plot $\delta(\varepsilon)$ from the expression above vs. ε (in log-log scale) as you vary ε between 0.1 and 10.

[9] **Hint**: We derived in class the set $S_\star$ that maximizes the right side of the hockey-stick divergence and the value of $P(S_\star)$. You need to calculate $Q(S_\star)$ similarly and plug it into $H_\alpha(P\|Q) = P(S_\star) - \alpha Q(S_\star)$.